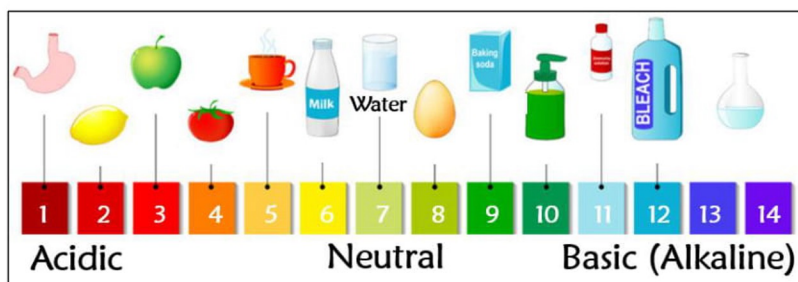


In a Nutshell: Ocean Acidification

Scientists are concerned about a trend called *ocean acidification*, a process that changes the chemical makeup of the ocean to make it more acidic. Every ecosystem is affected by any change, and oceans, covering over two-thirds of Earth's surface, contain an enormous amount of Earth's biodiversity.

What is pH?

pH is the measure of how acidic or basic a substance is. This is a scale from 0 to 14, with 7 being the neutral point. If the pH is closer to 0, the substance is more *acidic*. The closer it is to 14, the more

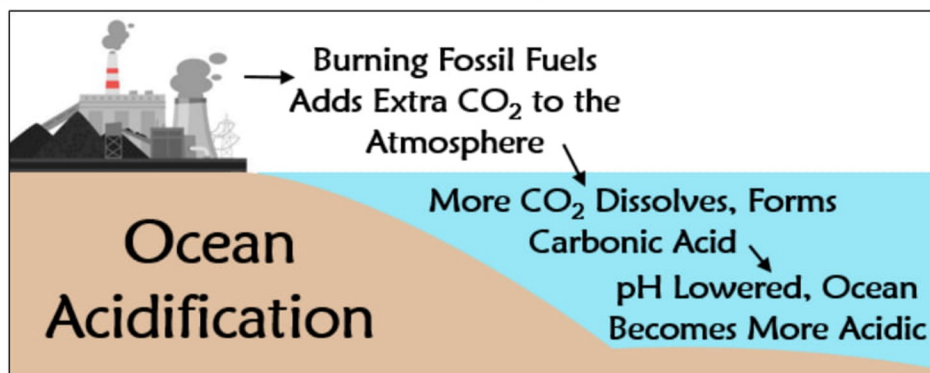


basic it is. Pure water measures 7.0 on a pH scale. Over the past 300 million years, ocean pH has been slightly basic, averaging about 8.2. Today, it is around 8.1, representing a 25% increase in acidity over the past 200 years. Even though this may not seem like a big difference, the effect on living organisms is significant. Lower ocean pH could cause major problems for marine organisms.

What is causing it?

The atmosphere is composed of gases including oxygen, nitrogen, and carbon dioxide (CO_2). The concentration of CO_2 in the atmosphere has increased since the Industrial Revolution. Humans burn more fossil fuels than ever before, which releases extra CO_2 into the atmosphere. Additionally, deforestation is killing enormous numbers of trees, which remove CO_2 from the atmosphere through the process of photosynthesis. So, between fossil fuels adding extra CO_2 and deforestation decreasing carbon-removers, the carbon dioxide levels in the atmosphere are higher than ever in Earth's history.

Ocean waters are constantly reacting with atmospheric gases. In fact, the oceans absorb up to half of the CO_2 in the atmosphere, acting as a *carbon sink*, a part of the environment that stores carbon. The upper layer of the ocean absorbs CO_2 . Here, the CO_2 reacts with carbonate ions and water to form carbonic acid. This is a very normal and natural process and helps keep carbon balanced in the *carbon cycle*, the process in which carbon is cycled through living and non-living things on Earth. However, because the level of CO_2 in the atmosphere is higher than ever before, the oceans are absorbing more CO_2 – and therefore more carbonic acid is being produced. This makes the ocean more acidic.



What are the effects?

As you might expect, a change in any ecosystem can easily lead to wide-ranging problems. Many different marine organisms are very susceptible to changing pH levels and a more acidic ocean. Below are some ways that ocean acidification impacts marine ecosystems:

- **Unable to Make Shells:** Because more carbonate ions are taken up in a reaction with carbon dioxide, there is less available for organisms that build their shells using carbonate ions, known as *calcifiers*. Organisms such as oysters, clams, sea urchins, corals, and calcareous plankton are all at risk when there are less carbonate ions.
- **Dissolving Shells:** Acidic water is *corrosive* to – it breaks down – the shells of shelled animals in the ocean, decreasing the thickness and strength of their shells.
- **Coral:** Acidification is already slowing coral growth rates and, along with increasing temperatures, may be leading to *coral bleaching*, a process where coral expel the algae living in them, making them white. This is a sign that corals are under stress.
- **Finfish:** Finfish such as herrings, sardines, tuna, cod, and many others are very important because, apart from their membership in marine food chains, they are also a source of food for humans. Ocean acidification will make them vulnerable as their habitats will change. It can also affect their behavior, *fitness* (ability to survive and reproduce) and survival of their larvae.
- **Biogeochemical Cycles:** CO₂ from the upper levels of the ocean is used by phytoplankton when they do photosynthesis as a part of a process called the *biological carbon pump*. Ultimately, the carbon goes deep into the ocean, which is how the ocean “stores” carbon. The efficiency of the pump depends on the amount of CO₂ absorbed by the ocean. Acidic water reduces the number of organisms that absorb CO₂, therefore reducing the ocean's capacity to perform its role as a carbon sink.



How can ocean acidification be reduced?

People are dependent on burning fossil fuels for transportation and electricity production. Switching to renewable energy, such as wind and solar, cuts down fossil fuel use, and therefore the amount of extra CO₂ entering the atmosphere. We can also conserve electricity, buy energy-efficient appliances, insulate our homes better, carpool, and other simple changes. Every decision to be energy-efficient leads to less CO₂ in the atmosphere.

The effects of ocean acidification could also be lessened if people implement strict policies to protect the oceans. If other stresses such as overfishing and pollution are reduced, the stress of acidification would be less profound.



Directions; Read the article *In a Nutshell: Ocean Acidification* and answer the questions below.

1. Use context clues from the text to define the following terms:

Ocean acidification – _____

pH – _____

Acidic – _____

Basic – _____

Carbon Sink – _____

Carbon Cycle – _____

Calcifiers – _____

Corrosive – _____

Coral Bleaching – _____

Fitness - _____

Biological Carbon Pump - _____

2. Describe the pH scale. What does it measure?

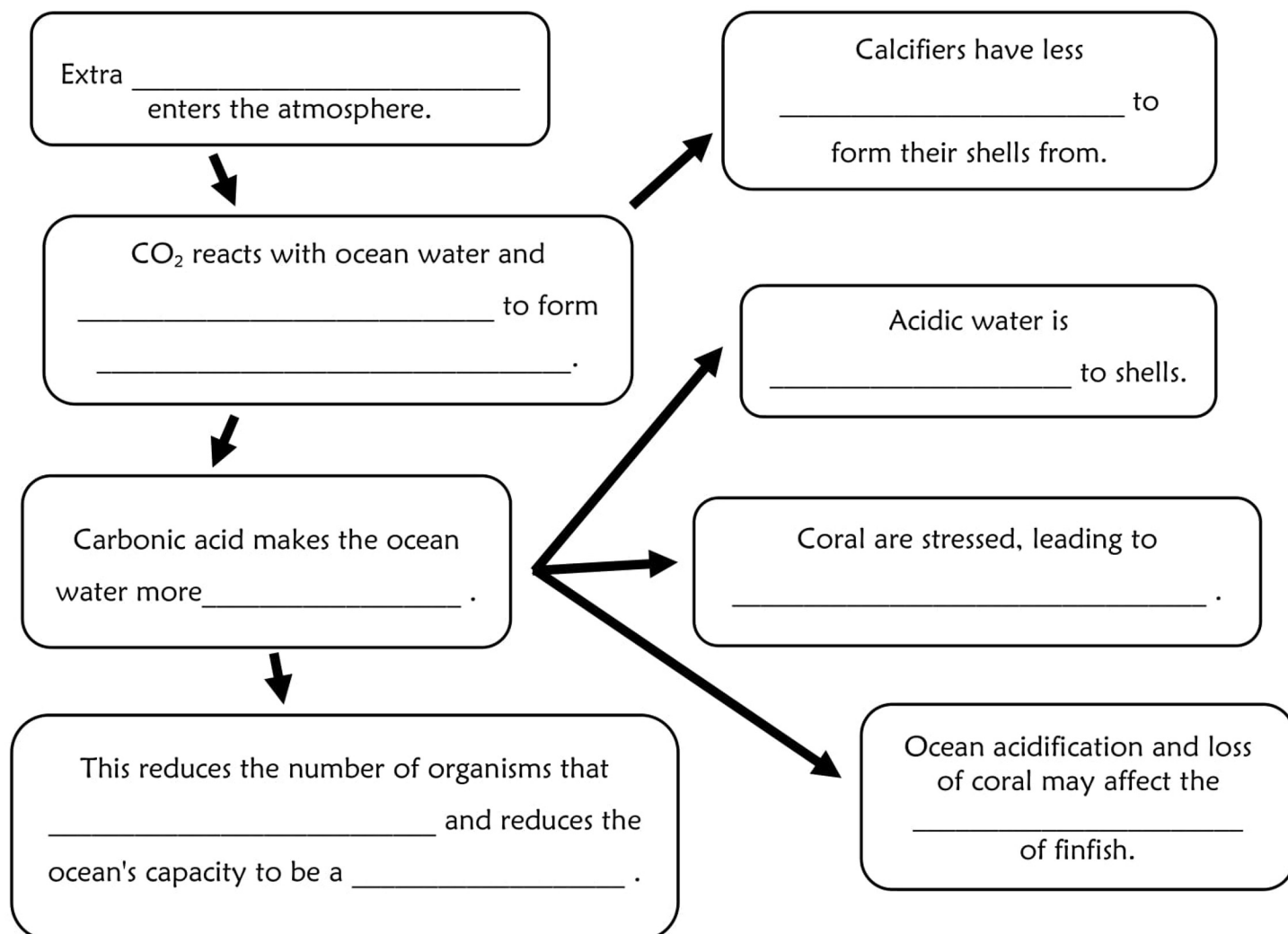
3. What had been the pH of Earth's oceans for most of Earth's history?

4. What is the pH of the oceans now? What percent increase in acidity does this represent?

5. Name and explain two things that contribute to extra carbon dioxide in our atmosphere.

6. Describe in words the chemical reaction of carbon dioxide that results in ocean acidification.

7. What are calcifiers? Name and describe two ways they are affected by extra carbon dioxide in the atmosphere.
8. How are coral affected by ocean acidification?
9. What makes finfish vulnerable due to ocean acidification?
10. How is carbon stored in the ocean?
11. What could be done to halt ocean acidification?
12. Complete the flow chart below to show how ocean acidification works and the effects it has.



Directions; Read the article *In a Nutshell: Ocean Acidification* and answer the questions below.

1. Use context clues from the text to define the following terms:

Ocean acidification – A reduction of PH in the ocean

pH – The level of acidity in the ocean

Acidic – Having a PH below 7

Basic – Having a PH closer to 14

Carbon Sink – Something that latkes in more carbon than it produces

Carbon Cycle – Reusing atoms which travel into the atmosphere

Calcifiers – Organisms the have shells

Corrosive – The acid that breaks down shells

Coral Bleaching – When a coral lets go of all of its algae

Fitness - The ability to survive and reproduce

Biological Carbon Pump - The carbon that goes deep into the ocean

2. Describe the pH scale. What does it measure?

A scale of 0-14 and it measures the acidification in water

3. What had been the pH of Earth's oceans for most of Earth's history?

6.5-7.7

4. What is the pH of the oceans now? What percent increase in acidity does this represent?

About 8.1

5. Name and explain two things that contribute to extra carbon dioxide in our atmosphere.

Deforestation and fossil fuels

6. Describe in words the chemical reaction of carbon dioxide that results in ocean acidification.

CO2

7. What are calcifiers? Name and describe two ways they are affected by extra carbon dioxide in the atmosphere.

When more carbon ions are taken up in a space a reaction with carbon with less available calcifer's that are building up their shell. It breaks down some of their shell by decreasing the strength and thickness of the shells.

8. How are coral affected by ocean acidification?

Coral bleaching which is when the coral releases its algae

9. What makes finfish vulnerable due to ocean acidification?

Changing their habitats, survival, fitness, and behavior

10. How is carbon stored in the ocean?

The CO₂ sinks down into the bottom of the ocean

11. What could be done to halt ocean acidification?

Stop deforestation and use less fossil fuels

12. Complete the flow chart below to show how ocean acidification works and the effects it has.

